🤖 MiniGPT Agent

A lightweight tool-augmented AI assistant built with FastAPI, Streamlit, and LLM integration (Grok API), featuring intent classification, tool routing, and memory support.

✨ Overview

MiniGPT Agent is an AI-powered assistant built using a tool-augmented LLM architecture. It performs intent classification to route user queries to specialized tools such as a calculator and web search, and then uses an LLM to generate context-aware responses.

It demonstrates modular AI agent design with:

Tool usage

Memory system API-based orchestration Frontend + backend separation 🚀 Features 🧠 Intent Classification Automatically detects query type (Math / Search / General) 🔎 Web Search Tool Mock search system (extensible to real APIs like SerpAPI/Tavily) 🧮 Safe Calculator Secure arithmetic evaluation using controlled logic 💬 LLM Integration Uses Grok API for natural language responses 💾 Memory System Stores chat history in JSON format 🌐 FastAPI Backend Handles agent pipeline and tool routing 🎨 Streamlit Frontend Clean chat-based UI for interaction

🧠 Architecture

User ↓ Streamlit UI ↓ FastAPI Backend ↓ Intent Classifier ↓ Tool Router ├── Calculator Tool ├── Web Search Tool └── LLM (Grok API) ↓ Memory Storage (JSON) ↓ Final Response

⚙️ Tech Stack

Python 🐍 FastAPI ⚡ Streamlit 🎨 Grok API / LLM 🤖 Pydantic 📦 Requests 🔗

📁 Project Structure

ai-agent-assistant/ │ ├── app/ │ ├── **init**.py │ ├── agent.py │ ├── intent.py │ ├── llm.py │ ├── memory.py │ └── tools/ │ ├── calculator.py │ └── search.py │ ├── app_ui.py ├── main.py ├── requirements.txt ├── chat_memory.json └── README.md

⚙️ Installation

1. Clone the repository git clone https://github.com/your-username/minigpt-agent.git cd minigpt-agent

2. Create virtual environment (optional but recommended) python -m venv venv source venv/bin/activate # Mac/Linux venv\Scripts\activate # Windows

3. Install dependencies

pip install -r requirements.txt

▶️ Running the Project

Start Backend (FastAPI) uvicorn main:app --reload Start Frontend (Streamlit) streamlit run app_ui.py

💡 Example Usage

Input: What is 25 * 90? Output: 2250

Input: Tell me about Artificial Intelligence Output: AI is a field of computer science focused on creating systems that simulate human intelligence...

🧩 Key Concepts Demonstrated

Tool-augmented LLM architecture Intent classification (rule-based routing) Modular backend design API communication between services State persistence using JSON memory Secure evaluation for math operations

⚠️ Notes

Calculator uses safe evaluation (no raw eval) Memory is stored locally in JSON file Grok API key required for LLM responses 👩‍💻 Author

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⭐ If you like this project

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